

Final Project

- Due Monday by 11:59p.m.
- Points 100
- Submitting a text entry box or a file upload
- Available until Apr 9 at 11:59p.m.

Note: This full assignment description was edited with help from Claude Opus 4.6.

Overview

This is the capstone project. You will analyze a real-world dataset using methods from across the course, and write up your findings as a coherent report.

Projects 1 and 2 built your skills in exploration, communication, and methodological rigour. Project 3 asks you to put it all together: choose the right methods for your data, use them to answer a clear question, and present the results as a unified narrative. The datasets are larger and messier than before — missing values, mixed variable types, thousands of observations — and part of the challenge is making sound preprocessing decisions.

Datasets

You will analyze **one** of the following four datasets, assigned via ranked-choice lottery.

Dataset	Domain	Observations	Features
NHANES	Health / Epidemiology	5,367	16
World Values Survey	Social Science	19,364	16
SDSS Stellar Classification	Astronomy	10,000	18
Open University Learning Analytics Education		25,820	15

NHANES (National Health and Nutrition Examination Survey): CDC survey data on US adults combining demographics, body measurements, blood pressure, and laboratory biomarkers. Missing data in several variables (~4–8%). No built-in class label, so the emphasis is on unsupervised methods, though you can derive groupings if you justify them.

World Values Survey (WVS): Responses from 19,364 people in 10 countries on religion, politics, trust, life satisfaction, and gender equality. Variables use different scales (binary, 4-point, 10-point), and negative values encode missing/refused responses. Country serves as a natural class label.

SDSS Stellar Classification: 10,000 celestial objects (stars, galaxies, quasars) from the Sloan Digital Sky Survey. Main features are five photometric magnitudes plus spectroscopic redshift. No missing data, but classes are imbalanced (50% galaxies, 42% stars, 9% quasars). Good for comparing unsupervised clustering against known labels.

Open University Learning Analytics (OULAD): 25,820 student-course registrations with demographics and assessment performance. Four outcome categories (Pass, Fail, Withdrawn, Distinction). Large dataset mixing categorical and continuous variables. Some missingness is informative (students with few assessments have no score variability).

Full documentation (variable descriptions, citations, loading code in Python/R/Julia) is available on Quercus.

What You Need to Do

Analyze your dataset, write a report, and submit reproducible code. Your report is evaluated on three things:

1. **Visualization & Communication (30%)** — Clear figures, good captions, readable writing with a coherent narrative arc.
2. **Methodological Rigour (30%)** — Justified choices, tested assumptions, sensitivity checks.
3. **Interpretation (40%)** — Meaningful findings that draw on multiple methods, connect to the domain, and build toward a clear conclusion.

Methods

The full course toolkit is available. Methods are organized into three tiers:

Baseline (expected in every submission):

- Exploratory Data Analysis (Week 1)
- MVN assessment and hypothesis testing (Weeks 2–3)
- PCA (Weeks 4–5)

General methods (use at least two):

Method	Weeks	When to use
CCA & Multivariate Regression	6	When your data has natural variable blocks or prediction targets
Factor Analysis	7	When latent continuous structure is plausible
ICA	8	When non-Gaussian independent sources are plausible
k-means & GMM	9	When cluster structure is plausible

Method	Weeks	When to use
LDA & Logistic Regression	9	When class labels are available
Extensions (use at least one):		
Method	Weeks	When to use
Robust estimation (MCD, outlier detection)	10	When outliers or contamination are concerns
Regularization & shrinkage	10	When p is large relative to n , or estimates are unstable, or you want to change the properties of the fit
Cross-validation	10	For honest error estimation and model/hyperparameter selection
Kernel methods (kernel PCA)	11	When linear methods miss nonlinear structure

Weeks 10–11 will be covered during the project window (by March 25). You can start your analysis with Weeks 1–9 methods and add extensions as you learn them. The extensions are not standalone analyses — they improve or stress-test your core analysis. For example, cross-validation gives you honest classification error rates, robust covariance tells you whether outliers are driving your PCA, and kernel PCA reveals whether linear projections are missing important structure.

Not every method applies to every dataset. If something doesn't fit, say so briefly and move on.

How to Approach the Analysis

Good analysis starts with exploration. Look at your data carefully — distributions, correlations, outliers, patterns — and let what you see guide your next steps.

The key habit is turning observations into questions. If your EDA shows two clusters in a scatter plot, the natural next step is “does GMM confirm this, and what are the clusters?” If you see heavy-tailed distributions, you might ask, “Does robust PCA change the component structure?” If you see high correlations among a subset of variables, you might ask, “Is there a latent factor?” Each method you apply should answer a question that your earlier analysis raised.

This is iterative. Methods generate new observations, which generate new questions, which motivate new methods. That's how real analysis works. A good project follows several of these question–method–answer chains and then connects them into a coherent picture.

Writing the Report

Your report should read like a short research article, not a lab notebook.

Start with a central question (or questions). Before you write, decide what your report is *about*. This should be substantive question(s) about your dataset that your analysis addresses. Examples:

- “What latent health profiles exist in the US adult population, and how do they relate to demographic factors?”
- “Do cultural value dimensions cut across national boundaries, or do countries form distinct value clusters?”
- “How well can photometric measurements alone recover spectroscopic classifications of celestial objects?”
- “What student engagement patterns predict academic outcomes, and can at-risk students be identified early?”

Your central question(s) give the report a through-line. Everything you include should connect back to it.

Structure your report as a narrative. A common structure:

1. What is the dataset? What is your central question? Why is it interesting?
2. **Exploratory analysis.** What does the data look like? What patterns, anomalies, or structures does initial exploration reveal? What questions does this raise?
3. **Methods and findings.** What methods did you apply and why? What did each one reveal? How do the results connect to each other?
4. What is the answer to your central question? Where did methods agree? Where did they disagree, and what does that tell you? What are the limitations?

The important thing is that the report has a *direction*. Each section should build on the previous one. The reader should finish understanding what you found and why it matters — not just what you ran.

A common pitfall is writing the report in the order you did the work. That usually produces a methods catalogue: “first we ran PCA, then GMM, then LDA, then kernel PCA.” Instead, organize around what you *learned*, using the methods as evidence.

Methods Scope

Use Weeks 1–11 material. Anything outside the course scope will be ignored during grading.

Submission Requirements

Submit the following on Quercus by **April 6, 2026, 11:59 PM**.

1. Written Report (PDF)

Length: 8–12 pages (excluding references)

Figures: 5–8 visualizations

Formatting:

- 11pt or 12pt font, 1-inch margins recommended

- Figures embedded near their discussion, with informative captions
- Page numbers
- References where appropriate

Page limits are enforced. Under 8 pages receives a score of 0. Content beyond page 12 is ignored (references excluded).

2. Code (Jupyter Notebook or Script)

Languages: Python (preferred), R, or Julia

Your code must be reproducible (runs with only path changes), organized (sections match the report), and commented where non-obvious.

3. AI Attribution (if applicable)

If you used generative AI tools, describe how. This can be a paragraph in your report, a separate file, or code comments. Failure to attribute AI assistance when used is an academic integrity violation.

Late Policy

10% penalty per day, maximum 3 days late. **After 3 days, submissions will not be accepted without accommodation.**

Evaluation

The rubric has three components:

Component	Weight	Sub-dimensions
Visualization & Communication	30%	Figure Design & Readability (15), Writing Quality & Narrative (15)
Methodological Rigor	30%	Method Selection & Justification (15), Assumption Testing & Robustness (15)
Interpretation of Structure	40%	Analysis Execution & Integration (20), Interpretation Depth & Synthesis (20)

Each sub-dimension is scored 0–4. A score of 4 is rare. Most competent submissions score 2–3. See the full rubric on Quercus.

Compared to P2, the weight shifts from rigour (40% in P2) to interpretation (40% in P3). Rigour is still expected — it's just no longer the primary focus.

FAQ

Do I need to use every method from the course?

No. You need EDA, PCA, at least two general methods, and at least one extension. Beyond that, use what makes sense for your data. A focused analysis of well-chosen methods is better than a shallow pass through everything.

What counts as a “general method” vs. an “extension”?

General methods are the core analytical techniques from Weeks 6–9 (FA, ICA, CCA, multivariate regression, k-means/GMM, LDA/logistic). Extensions are the tools from Weeks 10–11 that improve or validate your analysis (e.g., robust estimation, regularization, cross-validation, kernel methods). See the methods tables above.

What if Weeks 10–11 haven’t been covered yet when I start?

Start with Weeks 1–9. Weeks 10–11 will be taught by March 25. Plan to incorporate at least one extension in the later stages of your work.

What makes a good central question?

A good central question is specific to your dataset and answerable with the course methods. It should be something you genuinely want to know about the data, not just “what methods work?” Bad: “We analyze the NHANES dataset.” Better: “What latent health profiles exist among US adults, and do they differ by demographic group?”

What if my central question changes during the analysis?

That’s normal. You may start with one question and discover something more interesting. Your final report should present the question you ended up answering, even if it’s not where you started.

How should I handle missing data?

Make a deliberate choice and explain it. Common options: drop rows, drop variables, or impute. If your dataset has substantial missingness, checking whether conclusions hold under a different strategy is a good robustness check.

My dataset doesn’t have class labels. Do I need classification methods?

No. Focus on unsupervised methods. You can optionally use GMM-discovered clusters as pseudo-labels, but this is not required.

How many components/clusters/factors?

Use standard selection criteria (scree plots, parallel analysis, BIC, silhouette scores). When the criteria disagree, discuss the tradeoff. Interpretability matters.

What if I find something unexpected?

Report it. Anomalies, data quality issues, and surprising structures are worth discussing.

Getting Help

- **Office Hours:** See the syllabus for times and locations
- **Piazza:** Post questions (check if yours has been asked first)
- **Dataset questions:** Direct to your assigned Lead TA

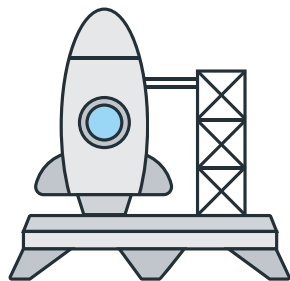
Start early. This is the largest project. The writing takes longer than you think.

File Upload

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Upload a file, or choose a file you've already uploaded.



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Choose a file to upload



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Submit Assignment

Project 3 Rubric

Criteria	Ratings						Pts
Code Reproducibility Your code must be reproducible — it should run with only minor path changes to the original data files.	0 pts Fail Code has significant errors, missing files, or cannot be reasonably executed			0 pts Pass Code runs or would run with trivial fixes			0 pts
AI Attribution If you used generative AI tools (ChatGPT, Claude, Copilot, etc.), you must provide attribution describing how the tools were used.	0 pts Fail AI clearly contributed but no attribution provided			0 pts No Marks Attribution provided (if AI was used) or no AI was used			0 pts
Collaboration Statement If work was done in substantial collaboration with others, you must provide a statement describing the nature of the collaboration	0 pts Fail Collaboration was not acknowledged even though it clearly occurred			0 pts No Marks Collaboration was outlined (if it occurred) or work was done independently			0 pts
Figure Design & Readability	15 pts Excellent	11 pts Good	7 pts Fair	4 pts Poor	0 pts Unacceptable		15 pts

Criteria	Ratings					Pts
Are the figures well-designed, easy to read, and accompanied by informative captions?	Publication-quality figures with informative captions that stand alone. Font sizes are appropriate throughout. Colours chosen thoughtfully. No overlapping text or clutter. Axes clearly labelled with units. Captions explain what the figure shows, highlight the key finding, and note why it matters. Would not look out of place in a journal article.	Well-designed figures with clear labels, legends, and readable fonts. Good colour choices. Captions explain what the figure shows and note key observations. Minor issues that don't impede understanding (e.g., slightly small text, one caption that describes but doesn't interpret).	Functional figures that convey information. Design is basic with some readability issues (small fonts, dense layouts). Default software styling with minimal customization. Captions describe what the figure shows, but don't highlight key findings.	Figures present but hard to read. Missing labels, legends, or titles. Poor colour choices. Captions are perfunctory labels only (e.g., "Figure 2: Factor loadings").	Missing figures, or figures so poorly designed that they fail to communicate. Missing or incorrect captions.	
Writing Quality & Narrative Does the report have a clear central question, a narrative arc where EDA motivates methods and findings build on each other, and writing that reads like a short research article rather	15 pts Excellent Reads like a short research article. Clear central question(s) are stated in the introduction and sustained throughout. EDA explicitly motivates method choices. Each section builds on the previous. Findings are organized	11 pts Good Central question(s) are present and mostly sustained. EDA connects to later methods. Clear organization with transitions between sections. The reader can follow the analytical thread. Some sections could	7 pts Fair Readable but feels like a methods catalogue. Data description is present, but EDA doesn't motivate method choices. No clear central question — the report describes what was done without saying why. Sections	4 pts Poor Difficult to follow. No central question. Minimal data introduction. Methods appear in arbitrary order with no motivation. Results listed without context.	0 pts Unacceptable Incoherent or severely disorganized. No data description. Major logical gaps.	15 pts

Criteria	Ratings					Pts
than a methods catalogue?	around what was learned, not what was run. The synthesis returns to the central question(s) and answers them. A reader unfamiliar with the dataset finishes with a clear understanding of what it contains and what the analysis revealed.	be better motivated or connected.	are loosely connected.			
Method Selection & Justification Are analytical choices explicitly justified? Does the student explain why they chose each method and why alternatives were or weren't appropriate?	20 pts Excellent Every major analytical choice is explicitly justified with clear reasoning grounded in the methods' assumptions and the properties of the data. Alternatives considered and their inclusion or exclusion explained. Methods chosen from multiple families with a clear rationale for how they complement each other.	15 pts Good Most analytical choices are justified. Explains why primary methods were selected. Key decisions are justified with at least one clear criterion. The reader understands why these methods were chosen for this data.	10 pts Fair Methods applied, but choices not well justified. May state what was done without explaining why. Methods seem chosen by default rather than by reasoning about the data.	5 pts Poor Methods applied with little or no justification. The reader cannot tell why these methods were selected over alternatives.	0 pts Unacceptable No justification for any choices. Methods are missing or applied in ways that suggest a fundamental misunderstanding.	20 pts

Criteria	Ratings					Pts
	Demonstrates understanding that method choice requires judgment, not just defaults.					
Assumption Testing & Robustness Are assumptions tested rather than assumed? Are results assessed for sensitivity to methodological choices?	20 pts Excellent Assumptions are systematically tested and results interpreted, not just reported. Distributional and structural assumptions are assessed with a clear discussion of violations and implications. At least one meaningful robustness or sensitivity check (varying key parameters, comparing preprocessing choices, handling missing data differently). Discusses how robust or fragile the findings are.	15 pts Good Key assumptions tested and interpreted. At least one robustness or sensitivity check was performed. Results interpreted, not just reported. Minor gaps in coverage.	10 pts Fair Some assumption testing present but perfunctory. Tests run, but results not interpreted. No robustness checks. Assumptions are treated as checkboxes.	5 pts Poor Minimal assumption testing. May mention assumptions in passing without formal tests. No robustness assessment.	0 pts Unacceptable No assumption testing. No evidence that the student considered whether methods are appropriate for this data.	20 pts
Analysis Execution &	15 pts Excellent Technically impeccable execution that	11 pts Good Correct execution that meets	7 pts Fair Analysis is present and mostly	4 pts Poor Analysis attempted but with notable	0 pts Unacceptable Analysis missing, fundamentally	15 pts

Criteria	Ratings					Pts
Integration Is the analysis technically correct? Does it meet the methods requirements (baseline + at least two general methods + at least one extension), and does it integrate methods rather than just running them in sequence?	exceeds the methodological requirements. Data appropriately preprocessed (standardization, missing data handled thoughtfully). Methods implemented correctly with proper settings. Baseline, general, and extension methods are all present, with their outputs connected. All relevant diagnostics computed. Numerical results are correct and consistent throughout the report.	the methods requirements: baseline methods (EDA, MVN assessment, PCA), at least two general methods, and at least one extension. Key outputs and diagnostics present. Some connection between the methods' results. Minor issues at most.	correct, but does not fully meet the methods requirements —may be missing general methods or extensions. Preprocessing adequate. Methods are applied independently without integration.	technical issues. May have errors in preprocessing or implementation. Limited to baseline methods only. Key outputs are missing.	flawed, or results clearly incorrect. Major conceptual errors.	
Interpretation Depth & Synthesis Does the analysis reveal meaningful structure and synthesize insights across methods?	15 pts Excellent Insightful synthesis that reveals genuine understanding. Latent dimensions, clusters, or discriminant features are given meaningful domain-specific names. Connects findings to	11 pts Good Solid interpretation and some synthesis. Structure is named meaningfully based on loading/feature patterns. Compares at least two methods and discuss the comparison. Makes connections	7 pts Fair Basic interpretation present. Reports the number of clusters/factors and lists important features, but interpretation is superficial. Methods run independently with results stated but not synthesized. No cross-	4 pts Poor Minimal interpretation. Reports numerical results without explaining their meaning. No domain connection. No synthesis. The reader learns what was computed but not what was discovered.	0 pts Unacceptable No interpretation, or an interpretation that is fundamentally wrong. Conclusions unsupported by the analysis.	15 pts

Criteria	Ratings					Pts
domain knowledge and earlier EDA. Compares what different methods reveal and discusses what each comparison tells us. Identifies where methods agree (convergent evidence) and where they disagree (revealing data complexity). The reader finishes understanding the data's structure from multiple perspectives. Discusses limitations and what the analysis cannot reveal.	between EDA and later analyses. The reader understands the key structures discovered.	method comparison.				
Total Points: 100						